Lesson 9.4 Teaching Support Guide

Teach-tomorrow route

Print: current Guided Notes and Practice A for the class; Exit Tickets; optional Language Support and supported homework. Keep this guide and answer keys teacher-facing. Use the full-size model pages in the updated notes.

Phase	Planned minutes	Teaching move
Retrieve	5	Warm-Up items 1–4; note the missing prerequisite before launch.
Launch	7	Name the session target and invite a prediction from the conceptual task.
Model	10	Link a visual model to the worked example; rehearse the mathematical explanation.
Practice	18	Core: A #1, #2, #4, #6 and the error task. Choose remaining items from evidence.
Check	5	Exit Ticket; use the diagnostic page before assigning reteaching.

This is a planned 45-minute route, not a measured classroom duration. For a shorter lesson, keep the model and exit; reduce repeated practice. Form B is parallel practice or reassessment only when its items are unseen.

S1: Substitute and Solve · I can substitute a value into an equation to find the matching output.

S2: Working Backward to a Goal · I can work backward from a goal to find the input an equation needs.

Spaced retrieval: revisit one item after 2 lessons and after approximately 1 week. Mix it with the next unit’s prerequisite check; do not announce the operation before students choose a method.

Session 1 · Substitute and Solve

I can substitute a value into an equation to find the matching output. · 6.EE.C.9

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 60 b. 23	Evaluate each term in its story role, then compare cost with the available amount.
2	14 × 35 = 490 boxes	Evaluate each term in its story role, then compare cost with the available amount.
3	B — 22	Ask the learner to show why B — 22 is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Adds the fee before multiplying or gives a value without answering the decision..

Response: Evaluate each term in its story role, then compare cost with the available amount.

Also check: Adding the input to the rate instead of multiplying (6m with m = 30 gives 36).

Also check: Adding the fee before multiplying ((14 + 25) × 6 instead of 14 × 6 + 25).

If calculation alone is wrong, use Session 1 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 1 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

A rental costs c=5h+9. Is $30 enough for 5 hours?

Teacher solution: Cost 34; no, it is $4 short.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: c=7(4)+12=40; exactly enough, assuming the given model has no other fees.

Misconception response anchor: Adds the fee before multiplying or gives a value without answering the decision.. Use evaluate each term in its story role, then compare cost with the available amount.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: The output ___ depends on ___ because the rule ___. Therefore ___.

Session 2 · Working Backward to a Goal

I can work backward from a goal to find the input an equation needs. · 6.EE.C.9

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. n = 12 b. n = 11	Write at least or at most first and test the neighboring whole-number outcomes.
2	11c = 528; c = 528 ÷ 11 = 48 cars	Write at least or at most first and test the neighboring whole-number outcomes.
3	B — subtracting 6	Ask the learner to show why B — subtracting 6 is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Always rounds up, including a maximum-budget problem..

Response: Write at least or at most first and test the neighboring whole-number outcomes.

Also check: Multiplying by the rate instead of dividing (4b = 96 answered as 384).

Also check: Dividing before subtracting the fee (600 ÷ 15 = 40 instead of 510 ÷ 15 = 34).

If calculation alone is wrong, use Session 2 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 2 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

At $8 each, find minimum items to raise $70 and maximum items affordable with $70.

Teacher solution: Minimum 9 ($72); maximum 8 ($64). Neighboring checks establish the constraints.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: Need 17: 16 gives $96, 17 gives $102. Can buy at most 16: 16 costs $96, 17 costs $102. Minimum-needed rounds up; maximum-affordable rounds down.

Misconception response anchor: Always rounds up, including a maximum-budget problem.. Use write at least or at most first and test the neighboring whole-number outcomes.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: The output ___ depends on ___ because the rule ___. Therefore ___.